

**Elementary Problems
Combinatorial Geometry
Solutions**

Version 1.0

10-12-2018

Basic Work

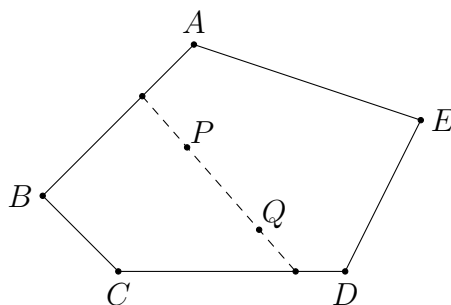
1. The answer is $\frac{n(n-1)}{2}$.

Each time when we draw the k th line, it forms at most $k-1$ intersection points with the lines which have been drawn as any two lines meet at no more than 1 point. By drawing a line which does not pass through any existing intersection point and is not parallel to any existing line, we can form exactly $k-1$ intersection points. It follows that at most

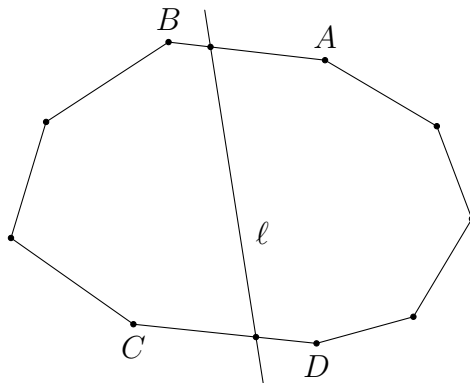
$$0 + 1 + 2 + \cdots + (n-1) = \frac{n(n-1)}{2}$$

intersection points can be formed.

2. Let the pentagon be $ABCDE$. The line PQ meets the pentagon at 2 points. Up to symmetry, these 2 points lie on AB and CD respectively, or AB and BC respectively. So P and Q must lie inside the quadrilateral $ABCD$.



3. It suffices to consider those lines ℓ cutting $\mathcal{P}$ in at least two points. Suppose ℓ cuts $\mathcal{P}$ at the sides AB and CD , possibly with $B = C$ or $A = D$ but not both. As $\mathcal{P}$ is convex, all other sides of $\mathcal{P}$ lie outside the convex quadrilateral (or triangle) $ABCD$, and hence they cannot meet the segment of ℓ lying inside $ABCD$. Also, since $\mathcal{P}$ is convex, the parts of ℓ lying outside $ABCD$ cannot intersect $\mathcal{P}$. Thus, ℓ meets $\mathcal{P}$ in at most two distinct points.

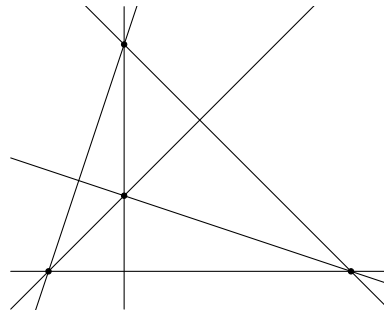


4. We prove by induction on n . The base case $n = 1$ is trivial. Suppose it holds for some n . When there are $n + 1$ red points and $n + 1$ blue points, there must be 2 consecutive points on the circumference which are of distinct colours, since otherwise all the points should have the same colour. We use these 2 points to form a chord so that it will not intersect any other chord formed by the remaining points. Then we apply the inductive hypothesis to the remaining $2n$ points. This gives a way to pair up these points to form n more chords with endpoints having distinct colours. So we are done by induction.
5. The answer is $n = 2, 3$ or $n \geq 6$. The case $n = 2$ trivially holds. When $n = 3$, we can take any 3 concurrent lines.

When $n = 4$, if there are 4 lines and 2 points such that at least 3 lines pass through each point, then there are 2 lines which pass through both points, which is impossible.

When $n = 5$, suppose there are 5 lines and 3 points such that at least 3 lines pass through each point. Let ℓ_1, ℓ_2, ℓ_3 be 3 concurrent lines. Then every pair of these lines should not meet at another point. WLOG we may assume ℓ_1, ℓ_4, ℓ_5 are concurrent at another point. But then there cannot be another group of 3 lines among the ℓ_j 's which could be concurrent at a third intersection point.

We shall prove by induction that there exists an admissible configuration for each of the cases $n \geq 6$. Indeed, we shall prove a stronger condition that there exist n straight lines and $n - 1$ points such that at least 3 lines pass through each of the $n - 2$ points and exactly 2 lines pass through the remaining point. For the base case $n = 6$, we can simply take a triangle together with its 3 altitudes. Then there are 3 lines which pass through each of the vertices and the orthocentre. Also, each foot of the altitude is the intersection of exactly 2 lines.



Suppose we have such a configuration for the case n . To construct the case $n + 1$, we simply need to draw a line ℓ which passes through the point which is the intersection of exactly 2 of the n lines. There are infinitely many choices for ℓ , and we can make a choice such that ℓ does not pass through any other existing intersection point and such that ℓ is not parallel to any existing line. Then there must be an intersection point formed by ℓ and another line. This point lies on exactly 2 lines. This establishes the inductive step and we are done.

6. The answer is $\frac{n^4 - 6n^3 + 23n^2 - 18n + 24}{24}$.

For $n = 1$, the number of region is 1. For $n = 2$, the number of regions is 2. For $n \geq 3$, to attain the maximal case, we may assume no three chords are concurrent.

Firstly, by joining the chords of consecutive points, we cut off a convex n -gon $A_1A_2 \cdots A_n$ and n small regions from the circle. Note that these small regions remain the same when all chords are joined. Thus, we may focus on the regions inside $A_1A_2 \cdots A_n$.

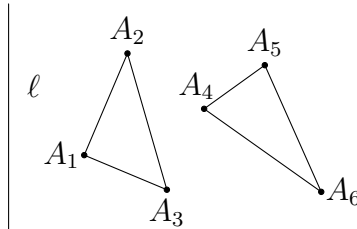
Secondly, we draw the diagonals A_iA_j one by one. When we draw each diagonal, it divides a region into 2. Thus, the number of regions is increased by 1. Every time when it meets another diagonal which has been drawn, it divides another region into 2. Again, the number of regions is increased by 1.

Therefore, the number of regions cut off by these diagonals is equal to the number of diagonals plus the number of intersection points plus 1 (which corresponds to the convex n -gon). For each vertex A_i , there are $n - 3$ diagonals joining to it. But each diagonal is counted twice in this way. So there are $\frac{n(n-3)}{2}$ diagonals in total. For the intersection point, each of them must be formed by the diagonals of a convex quadrilateral, and every such quadrilateral corresponds to exactly one intersection point. So there are $\binom{n}{4}$ intersection points.

Hence, the maximal number of regions is

$$\begin{aligned} n + \frac{n(n-3)}{2} + \binom{n}{4} + 1 &= n + \frac{n(n-3)}{2} + \frac{n(n-1)(n-2)(n-3)}{24} + 1 \\ &= \frac{n^4 - 6n^3 + 23n^2 - 18n + 24}{24}. \end{aligned}$$

7. We choose a line ℓ which is not parallel to any line formed by joining 2 of the given points. WLOG we may assume ℓ is a vertical line, and the given points $A_1, A_2, \dots, A_{3n}$ are ordered from left to right. Then we form a triangle using the points $A_{3k-2}, A_{3k-1}, A_{3k}$ for $k = 1, 2, \dots, n$. For each such triangle, it lies between the two lines passing through point A_{3k-2} and point A_{3k} respectively which are parallel to ℓ . Thus, there are no other points inside the triangle.



8. Let $\mathcal{P} = A_1A_2 \cdots A_{2n}$. For each of the $2n$ sides, say A_nA_{n+1} , we claim that there are at most $n - 2$ diagonals parallel to it. Clearly, no two diagonals parallel to A_nA_{n+1}

may share a common vertex. Since there are $2n - 2$ vertices other than A_n, A_{n+1} , at most $\frac{2n-2}{2} = n - 1$ diagonals parallel to $A_n A_{n+1}$ can be formed. If all these $2n - 2$ vertices form exactly $n - 1$ parallel diagonals, then there is a unique way to form the diagonals since no two of them as well as $A_n A_{n+1}$ can intersect. Indeed, we must have $A_n A_{n+1} // A_{n-1} A_{n+2} // \cdots // A_1 A_{2n}$. However, $A_1 A_{2n}$ is indeed a side of $\mathcal{P}$ instead of a diagonal. This is a contradiction. So at most $n - 2$ diagonals are parallel to $A_n A_{n+1}$.

The same bound holds for any of the sides. Thus, at most $2n(n - 2)$ diagonals may be parallel to one of the sides. However, there are

$$\binom{2n}{2} - 2n = n(2n - 1) - 2n = 2n^2 - 3n > 2n(n - 2)$$

diagonals in total. So there exists one of them which is not parallel to any side of $\mathcal{P}$.

9. The result holds when $n = 1$. Suppose $n \geq 2$. If $\triangle ABC$ is isosceles with $AB = AC$, then A lies on the perpendicular bisector of BC . For each pair of points B, C , there are at most 2 such points A on the perpendicular bisector of BC since no 3 points are collinear. Hence, there are at most $2 \binom{n^3}{2} = n^3(n^3 - 1)$ isosceles triangles formed among the n^3 points.

Note that each isosceles triangle is included in $\binom{n^3 - 3}{n - 3}$ groups of n points. Suppose on the contrary that any group of n points contains an isosceles triangle. Then the number of isosceles triangles is at least

$$\binom{n^3 - 3}{n - 3}^{-1} \binom{n^3}{n} = \frac{(n - 3)!(n^3 - n)!}{(n^3 - 3)!} \cdot \frac{(n^3)!}{n!(n^3 - n)!} = \frac{n^3(n^3 - 1)(n^3 - 2)}{n(n - 1)(n - 2)}.$$

We now prove that $n^3 - 2 > n(n - 1)(n - 2)$ for $n \geq 2$. This is equivalent to $3n^2 > 2n + 2$, which holds since

$$3n^2 \geq 6n = 2n + 4n > 2n + 2$$

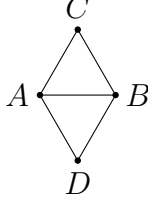
for $n \geq 2$. Therefore, the number of isosceles triangles is greater than $n^3(n^3 - 1)$, which is a contradiction. So there must be n points which do not contain any isosceles triangle.

10. Suppose there is no such pair of points. Consider any equilateral $\triangle ABC$ of side length 1. Since every two vertices have distance 1, they have pairwise distinct colours. WLOG assume A, B, C are of colour a, b, c respectively. Let D be a point distinct from C such that $\triangle BAD$ is equilateral. Then D has colour different from A and B , and hence D has colour c . The points C and D have distance $\sqrt{3}$, and they have the same colour.

Conversely, for any point D at a distance $\sqrt{3}$ from C , we can take the corresponding points A and B such that $\triangle ABC$ and $\triangle BAD$ are equilateral and have side length 1. The above arguments then show that C and D have the same colour. Thus, all points

on the circle with centre C and radius $\sqrt{3}$ have colour c . It is obvious that there are two points on this circle which have distance 1. This is a contradiction.

Therefore, there exist two points of the same colour whose distance is 1.



11. Suppose there exists a horizontal line consisting of exactly two colours. WLOG assume there exists $a \in \mathbb{Z}$ such that each (x, a) is red or blue. Let (b, c) be any green point. WLOG assume (b, a) is red. Then we choose any blue point (d, a) . This gives a right-angled triangle formed by the points (b, c) , (b, a) , (d, a) having distinct colours.

Suppose there exists a horizontal line consisting of only one colour. WLOG assume there exists $a \in \mathbb{Z}$ such that each (x, a) is red. Let (b, c) be any blue point. If there exists a green point (d, c) in the horizontal line $y = c$, then the right-angled triangle (b, a) , (b, c) , (d, c) has vertices with distinct colours. If the horizontal line $y = c$ only consists of red and blue points, then we are done from the first case. So it remains to consider the case when the horizontal line $y = c$ is entirely blue. Similarly, there exists a green horizontal line $y = e$. We can find a right-angled triangle with vertices lying on $y = a$, $y = c$ and $y = e$ respectively. Indeed, we claim that (e, a) , (c, c) , $(2c - a, e)$ form a right-angled triangle. This amounts to check

$$\begin{aligned} & \left(\sqrt{(c-e)^2 + (c-a)^2} \right)^2 + \left(\sqrt{(2c-a-c)^2 + (e-c)^2} \right)^2 \\ &= \left(\sqrt{(2c-a-e)^2 + (e-a)^2} \right)^2. \end{aligned}$$

Now, we may assume every horizontal line consists of three distinct colours. Similarly, we may assume each vertical line consists of three distinct colours. Let (a, b) be any red point. There exists a blue point (c, b) in the same horizontal line, and a green point (a, d) in the same vertical line. They clearly form a right-angled triangle.

Therefore, we are done in every case.

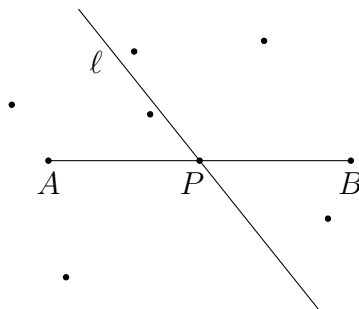
12. The answer is n . If the set contains the vertices of a regular polygon with n sides, then this set has n axes of symmetry in total.

Now, suppose a set of n points has at least $n + 1$ axes of symmetry. Note that the centroid G of the n points remains unchanged under the reflection in each axis of symmetry. Therefore, all axes of symmetry pass through G . Let P be one of the n points. Consider its image under the $n + 1$ reflections. For any point $Q \neq P$, there is at

most one reflection which maps P to Q , which is the perpendicular bisector of PQ . As there are only $n - 1$ such points Q , there must be 2 reflections which map P to itself. So P lies on the intersection of these axes of symmetry, which means $P = G$. As this holds for any point P , this yields the contradiction that all points are the same.

Therefore, there are at most n axes of symmetry.

13. There are finitely many perpendicular bisectors formed by pairs of the $2n$ points. So there exists a point P on the plane which does not lie on any of the perpendicular bisectors. We draw a circle with centre P which includes all the $2n$ points in its interior. We reduce the size of the circle gradually. Note that the circle cannot meet 2 of the $2n$ points simultaneously as P does not lie on the perpendicular bisectors. Hence, after meeting exactly n points, the other n points lie in the interior of the circle at that moment.
14. Let P be an arbitrary point on the segment AB such that P does not lie on any other line joining two points in $\mathcal{A}$. Such a point exists since there are finitely many points in $\mathcal{A}$. Initially, let ℓ be the line AB . We rotate ℓ anticlockwise with centre P and an angle slightly larger than 180° . Then ℓ always divides the plane into two regions $\mathcal{R}_1$ and $\mathcal{R}_2$. When ℓ rotates, we identify the two regions such that each region changes continuously.



Let N be the number of points in region $\mathcal{R}_1$. When ℓ has just started to rotate, the value of N is N_0 . At the end, ℓ has just crossed the line AB again, but with different orientations. This means the regions $\mathcal{R}_1$ and $\mathcal{R}_2$ are swapped. Thus, the value of N is $2n - N_0$. Each time when ℓ sweeps across a point in $\mathcal{A}$, such a point is uniquely determined from the choice of P . So the value of N is increased or decreased by exactly 1. To change from N_0 to $2n - N_0$, since either $N_0 \leq n \leq 2n - N_0$ or $2n - N_0 \leq n \leq N_0$, there must be a situation for which $N = n$. We take the line ℓ in that situation. Then there are n points in region $\mathcal{R}_1$ as well as n points in region $\mathcal{R}_2$. Also, A and B must be separated by ℓ . So we are done.

15. The answer is 9.

We use points to denote the birds. Let Γ be a circle with the greatest number of points. We first show that Γ has at least 5 points. If this is not true, then any circle contains

at most 4 points. There are $\binom{10}{5} = 252$ groups of 5 points, and each group consists of 4 concyclic points. On the other hand, each group of 4 concyclic points belong to 6 groups of 5 points. So there are $\frac{252}{6} = 42$ different circles each consists of 4 points.

By the pigeonhole principle, there exists a point A lying on at least $\left\lceil \frac{42 \times 4}{10} \right\rceil = 17$ circles. Each such circle contains 3 more points. Again by the pigeonhole principle, there exists another point B lying on at least $\left\lceil \frac{17 \times 3}{9} \right\rceil = 6$ of these circles. Then by the pigeonhole principle, there exists a third point C lying on at least $\left\lceil \frac{6 \times 2}{8} \right\rceil = 2$ of these circles. This gives two distinct circles passing through all of A, B, C , which is impossible.

Therefore, Γ has at least 5 points A, B, C, D, E . Consider 2 other points F and G not on Γ (if exist). By considering A, B, C, F, G , WLOG we may assume A, B, F, G are concyclic. By considering C, D, E, F, G , we may assume C, D, F, G are concyclic. Now, consider A, C, E, F, G . If A, X, F, G are concyclic where $X = C, E$, then since A, B, F, G are concyclic, we know that A, B, X, F, G are concyclic. However, the circumcircle of $\triangle ABX$ is Γ . This shows F, G lie on Γ , which is a contradiction. Similarly, if C, E, F, G are concyclic, then C, D, E, F, G are concyclic and hence we obtain a contradiction. So Γ contains at least 9 points.

Conversely, if there are 9 points lying on the same circle, then the condition is clearly satisfied. So the minimum number of birds on Γ is 9.

Geometric Facts

16. For each diameter of the circle, we colour one of the endpoint red and the other blue. Since any right-angled triangle inscribed in the circle must have 2 vertices being the endpoints of a diameter, its vertices cannot be of the same colour.
17. The answer is $\frac{n(n-1)(n-3)}{8}$ if n is odd, and is $\frac{n(n-2)(n-4)}{8}$ if n is even.

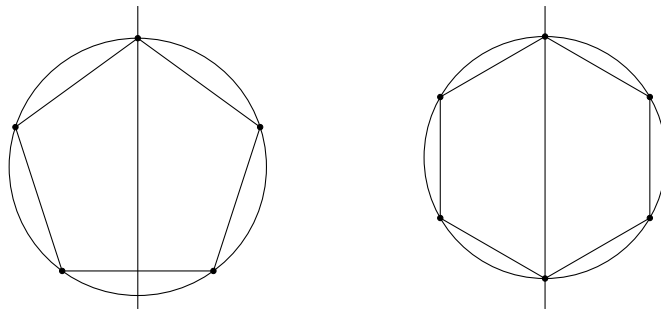
We count those obtuse triangles having A_1 as one of the vertices such that A_1 is not the obtuse angle. Let ℓ be the line passing through A_1 and the centre of the circumscribed circle of the regular n -gon. Then $\triangle A_1 A_i A_j$ is obtuse with A_i or A_j being the obtuse angle if and only if A_i and A_j lie on the same side of ℓ .

For odd n , there are $\frac{n-1}{2}$ vertices on each side of ℓ . Thus, there are

$$2 \binom{\frac{n-1}{2}}{2} = \frac{(n-1)(n-3)}{4}$$

such triangles. The same holds for any other vertex instead of A_1 . However, every obtuse triangle is counted twice in this way. Therefore, the number of obtuse triangles is

$$\frac{n(n-1)(n-3)}{8}.$$



Similarly, for even n , there are $\frac{n-2}{2}$ vertices on each side of ℓ . In total, there are

$$\frac{n}{2} \cdot 2 \binom{\frac{n-2}{2}}{2} = \frac{n(n-2)(n-4)}{8}$$

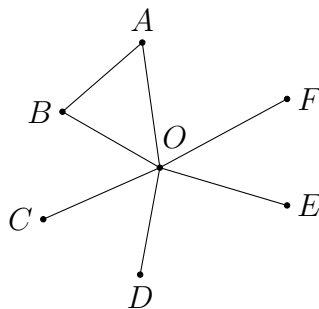
obtuse triangles.

18. Suppose there are k vertices of the triangles which lie in the interior of the polygon. We find the sum of interior angles of all triangles. Firstly, since there are m triangles, the sum is clearly $m \cdot 180^\circ$. Secondly, each interior point contributes 360° while the vertices of the polygon together contribute $(n-2) \times 180^\circ$. Therefore, we have

$$180m = 360k + 180(n-2),$$

which implies $m + n = 2k - 2 + 2n$. This shows $m + n$ is even.

19. Suppose on the contrary that there exists a point O which is the closest point to at least 6 points A, B, C, D, E, F .



Consider the (directed) angles $\angle AOB, \angle BOC, \angle COD, \angle DOE, \angle EOF, \angle FOA$. By the pigeonhole principle, there exists one of the angles, say $\angle AOB$, which is at most $\frac{360^\circ}{6} = 60^\circ$. In $\triangle OAB$, the largest interior angle is at least 60° . As it cannot be an equilateral triangle, this largest angle must be greater than 60° . Let $\angle OBA > 60^\circ$. Then $AO > AB$, which is a contradiction since we have assumed the closest point to A is O .

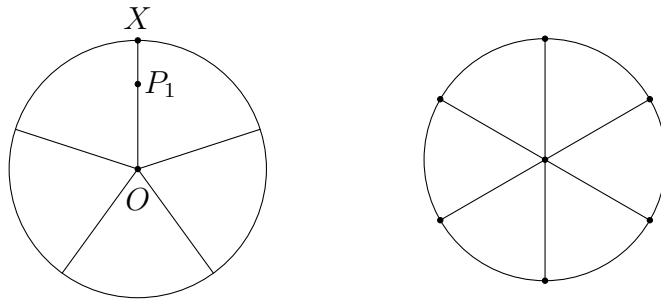
20. Firstly, consider $n = 2, 3, 4, 5, 6$. We choose a point X on the boundary of the disk so that P_1 lies on the segment joining X and the centre O of the disk. Consider the regular n -gon on the circle with X as one of the vertices. Its vertices together with O divides the disk into n equal sectors. Since P_1 belongs to the boundary of two sectors, two of the points P_i, P_j lie in the same sector (or on the boundary) by the pigeonhole principle. The longest length of a segment lying in a sector can be the longest chord or the radius. Since $n \leq 6$, the chord is longer than the radius. Thus, by the cosine formula, we have

$$P_i P_j \leq \sqrt{1^2 + 1^2 - 2 \cdot 1 \cdot 1 \cdot \cos \frac{360^\circ}{n}} = \sqrt{2 - 2 \cos \frac{360^\circ}{n}}.$$

It is possible that all the distances are not less than this upper bound value. For example, it is attained when $P_1 P_2 \cdots P_n$ form a regular n -gon on the circle. Thus, we find that

$$D_2 = 2, D_3 = \sqrt{3}, D_4 = \sqrt{2}, D_5 = \sqrt{\frac{5 - \sqrt{5}}{2}}, D_6 = 1.$$

Next, for $n = 7$, as there are more than 6 points, it is obvious that $D_7 \leq D_6 = 1$. Indeed, we have $D_7 = 1$. The optimal case can be attained when 6 of the points form a regular hexagon on the circle and the remaining point is the centre of the disk.



21. No. Since there are $\binom{5}{2} = 10$ pairs of points, their distances are exactly $1, 2, \dots, 10$ respectively. Suppose points A and B have distance 1. For any other point C , WLOG assume $AC > BC$. Then $AC \geq BC + 1$. By the triangle inequality, we need

$$AC \leq BC + AB = BC + 1.$$

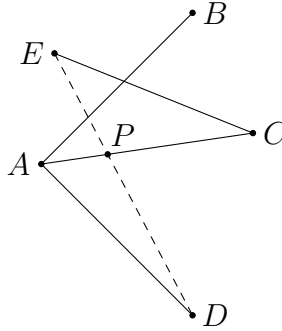
Therefore, equality holds and C lies on the line AB . As this is true for every other point, all 5 points lie on the same line.

Suppose A, B, C, D, E lie in that order on a line. Then the sum of the 10 distances are $4AB + 6BC + 6CD + 4DE$, which is an even number. However, $1 + 2 + \dots + 10 = 55$ is odd. This is a contradiction. Therefore, no such 5 points exist.

22. We prove by induction on n . When $n = 3$, there are only 3 segments. Thus, the number of diameters is at most 3. Assume the number of diameters of any set of k points is at most k for some $k \geq 3$. Consider any set of $k + 1$ points.

Suppose on the contrary that there are at least $k + 2$ diameters. If there is a point which is the endpoint of at most 1 diameter, we remove this point and consider the remaining k points. Since we have removed at most 1 diameter, there are at least $k + 1$ diameters remaining. This contradicts the inductive hypothesis.

Therefore, we may assume every point is the endpoint of at least 2 diameters. As each diameter corresponds to 2 endpoints, there are $\left\lceil \frac{2(k+2)}{k+1} \right\rceil = 3$ diameters having a common endpoint A by the pigeonhole principle. Suppose AB, AC, AD are diameters. WLOG we may assume B and D lie on different sides of the line AC as shown.



As C is the endpoint of at least 2 diameters, there must be another diameter CE . Note that CE cannot intersect AB and AD simultaneously. WLOG assume CE and AD do not intersect. (Here, E could be the point B .) Let P be the intersection of AC and DE , which may not be one of the $k + 1$ points. By the triangle inequality, we have

$$CE + AD \leq (CP + PE) + (DP + PA) = (CP + PA) + (PE + DP) = CA + ED.$$

As CE and AD are diameters, the lengths CA, ED cannot exceed CE, AD . Thus, equality must hold. This means A, C, D, E are collinear. However, for CE, AD, CA, ED to have the same length, one finds that some of A, C, D, E must coincide. This is a contradiction. Therefore, there are at most $k + 1$ diameters. The general case follows by induction.

23. Let $r = \frac{n+1}{n-1}$. Suppose on the contrary that $\frac{AB}{AC} > r$ whenever $AB \geq AC$. Let A_1, A_2 be 2 points such that A_1A_2 is maximized. WLOG assume $A_1A_2 = 1$. Suppose $A_1A_2 \geq A_1A_3 \geq \dots \geq A_1A_n$. From our assumption, we have $A_1A_k < \frac{1}{r^{k-2}}$ for each $k \geq 2$. The same set of upper bounds holds for the lengths $A_2A_3, A_2A_4, \dots, A_2A_n$, possibly with the indices being permuted.

For odd n , consider $k = \frac{n+3}{2}, \frac{n+5}{2}, \dots, n$. Then we have $A_1A_k \leq r^{-\frac{n-1}{2}}$. Among these indices, at most $\frac{n-3}{2}$ of them satisfy $A_2A_k \geq r^{-\frac{n-1}{2}}$ from the above bounds. Hence, there exists k such that $A_2A_k < r^{-\frac{n-1}{2}}$. By the triangle inequality, we have

$$2r^{-\frac{n-1}{2}} > A_1A_k + A_2A_k \geq A_1A_2 = 1.$$

This implies $r^{n-1} < 4$. However, we have

$$\begin{aligned} \left(\frac{n+1}{n-1}\right)^{n-1} &= \left(1 + \frac{2}{n-1}\right)^{n-1} \\ &\geq 1 + \binom{n-1}{1} \frac{2}{n-1} + \binom{n-1}{2} \left(\frac{2}{n-1}\right)^2 \\ &= 1 + 2 + \frac{2(n-2)}{n-1} \geq 4 \end{aligned}$$

as $n \geq 3$. This is a contradiction.

Similarly, for even n , consider $k = \frac{n+4}{2}, \frac{n+6}{2}, \dots, n$. Then we have $A_1A_k \leq r^{-\frac{n}{2}}$. Among these indices, at most $\frac{n-4}{2}$ of them satisfy $A_2A_k \geq r^{-\frac{n-2}{2}}$ from the above bounds. Hence, there exists k such that $A_2A_k < r^{-\frac{n-2}{2}}$. By the triangle inequality, we have

$$r^{-\frac{n}{2}} + r^{-\frac{n-2}{2}} > A_1A_k + A_2A_k \geq A_1A_2 = 1.$$

This implies $r^n < (r+1)^2$. As in the odd case, we have

$$r^{n-1} \geq 1 + 2 + \frac{2(n-2)}{n-1} = \frac{5n-7}{n-1}.$$

So we get $(r+1)^2 > \frac{5n-7}{n-1}r$, i.e. $\frac{4n^2}{(n-1)^2} > \frac{(5n-7)(n+1)}{(n-1)^2}$. This is the same as $n^2 < 2n+7$. This is impossible for $n \geq 4$ since

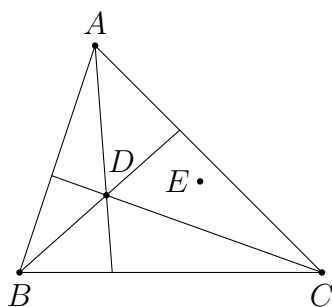
$$n^2 \geq 4n = 2n + 2n > 2n + 7.$$

Hence, there must be three points A, B, C satisfying $1 \leq \frac{AB}{AC} \leq \frac{n+1}{n-1}$.

Convex Hull

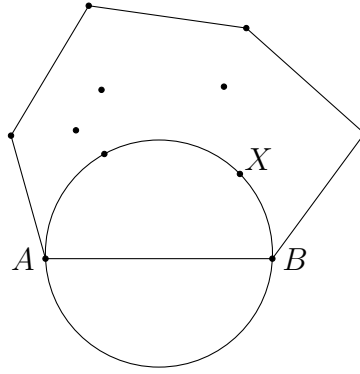
24. Consider the convex hull of the 5 points, if it is a pentagon or a quadrilateral, then we are done by choosing 4 vertices of the convex hull. If the convex hull is a triangle ABC , then the other two points D, E lie inside $\triangle ABC$.

After fixing the point D , the lines AD, BD, CD divide the triangle into 6 regions. WLOG assume E lies inside $\triangle ACD$. Also, we may assume E and C lie on the same side of the line BD . Then it is clear that $BCED$ forms a convex quadrilateral. Thus, we are done in every case.



25. Suppose such a function exists. Consider any convex pentagon and let the images of its vertices be A, B, C, D, E . Consider the convex hull of $ABCDE$. If the convex hull is a pentagon $ABCDE$ or a quadrilateral $ABCD$, then $ABCD$ is convex, which contradicts the property of the function. If the convex hull is a triangle ABC , then D and E lie inside $\triangle ABC$. WLOG assume E lies inside $\triangle ACD$. Also, we may assume E and C lie on the same side of the line BD . Then $BCED$ is a convex quadrilateral, which is again a contradiction. Thus, there is no such function.
26. Since any 4 points form a quadrilateral, no 3 points can be collinear. Consider the convex hull $A_1A_2 \cdots A_k$ of the n points. This convex hull can be partitioned into the triangles $\triangle A_1A_2A_3, \triangle A_1A_3A_4, \dots, \triangle A_1A_{k-1}A_k$. If there exists a point B inside the convex hull, then it must lie in some $\triangle A_1A_jA_{j+1}$ where $2 \leq j \leq k-1$. But then A_1, A_j, A_{j+1}, B do not form a convex quadrilateral. So no such point B exists. In other words, the convex hull is a convex polygon with n sides, whose vertices are the n points.

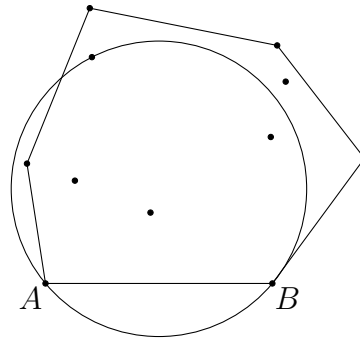
27. Yes. Let AB be a side of the convex hull of the points. Then the other points lie on the same side of the line AB . Choose a point X among the other $n - 2$ points such that $\angle AXB$ is maximized. Then any other point P must lie on or outside (AXB) since $\angle APB \leq \angle AXB$. Thus, this choice is admissible.



28. Let AB be a side of the convex hull of the points. Then the other points lie on the same side of the line AB . Denote the other points by $P_1, P_2, \dots, P_{2n+1}$ such that

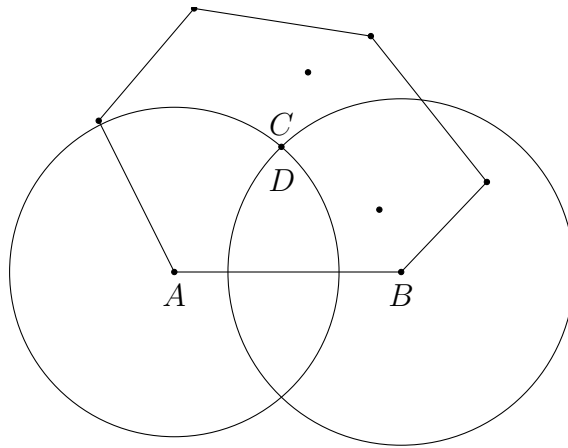
$$\angle AP_1B \leq \angle AP_2B \leq \dots \leq \angle AP_{2n+1}B.$$

Since no four points are concyclic, all these inequalities must be strict inequalities. Therefore, if we choose the middle point P_{n+1} , then $(AP_{n+1}B)$ will contain n points (which are $P_{n+2}, P_{n+3}, \dots, P_{2n+1}$) in its interior while the other n points (which are $P_1, P_2, \dots, P_n$) will lie outside.

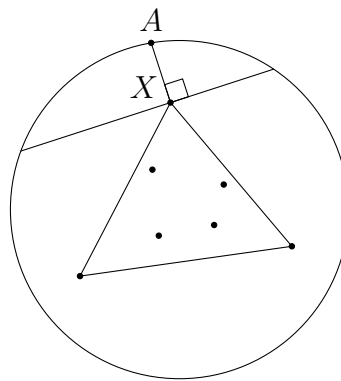


Let n be a positive integer. There are $2n + 3$ points on the plane with no three collinear and no four concyclic. Prove that there exists 3 of the points whose circumcircle splits the other $2n$ points into two halves, with n points lying inside the circle and n points lying outside the circle.

29. Suppose on the contrary that $n \geq k^2 + 3$. Let AB be a side of the convex hull of the points. For any of the other $n - 2$ points C , since there are only $k^2 < n - 2$ possibilities of the pairs (AC, BC) of lengths, there exist two points C and D such that $AC = AD$ and $BC = BD$ by the pigeonhole principle. This shows both C and D lie on the circle with centre A and radius AC and the circle with centre B and radius BC . However, this implies $C = D$ since C and D lie on the same side of the line AB . This is a contradiction. Thus, we must have $n \leq k^2 + 2$.

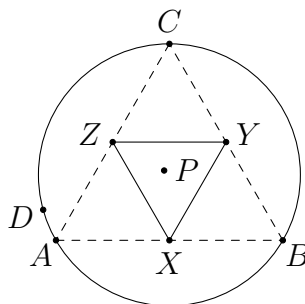


30. (a) Consider the convex hull of S . We first show that every vertex of this convex hull can be a closest point to a point on the circle. Let X be such a vertex. Then there exists a straight line ℓ passing through X which does not intersect the convex hull. We draw the perpendicular from X to ℓ . It meets the circle at two points, one of which is a point A which lies on different sides of ℓ as the convex hull. Then X must be the closest point in S to A .



If the convex hull has at least 3 vertices, then we are done. Otherwise, all the points in S must lie on the same straight line m . Then we simply draw a perpendicular to m from each point P in S . It meets the circle at a point whose closest point in S is P . So every point in S satisfies the property. In particular, there are 3 of them.

- (b) Let $\triangle ABC$ be an equilateral triangle on the circle. Let X, Y, Z be the midpoints of AB, BC, CA respectively. Suppose S contains X, Y, Z and $n - 3$ more points which are close to the centre of the circle. Let P be one of these $n - 3$ points. Consider any point D on the circle. Then one of DX, DY, DZ does not exceed the length AX , which is $\frac{\sqrt{3}}{2}$ times the radius of the circle. So this distance is less than DP . Thus, the closest point in S to D must be one of X, Y, Z . There cannot be 4 points in S satisfying the condition.



31. The answer is $3n - 6$. Let the convex hull be a polygon with k sides. Then there are $n - k$ points lying inside the convex hull. Consider any way of drawing the segments such that the number of segments is maximized. Then all the sides of the convex hull must be drawn since they do not intersect all other segments joining two of the points. Also, the convex hull is dissected into triangles by the segments drawn, since otherwise we can draw more diagonals of a polygon with at least 4 sides without intersecting any other segments.

Suppose there are m triangles. Then the sum of interior angles of all triangles is $m \cdot 180^\circ$. Also, each interior point contributes 360° while the vertices of the convex hull together contribute $(k - 2) \times 180^\circ$. Therefore, we have

$$180m = 360(n - k) + 180(k - 2),$$

which implies $m = 2n - k - 2$. The triangles consist of $3m$ sides in total. Each segment inside the convex hull contributes 2, while each side of the convex hull contributes 1. Suppose there are t segments in total, where $t - k$ of them are inside the convex hull. Then we have

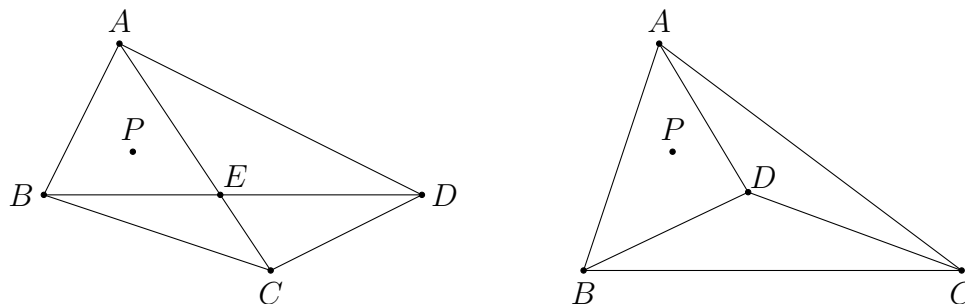
$$3(2n - k - 2) = 2(t - k) + k,$$

and hence $t = 3n - k - 3$. As $k \geq 3$, we have $t \leq 3n - 6$. Equality holds when $k = 3$, i.e. the convex hull is a triangle.

32. We first show that the assertion holds when there are 4 points A, B, C, D . We are done if P does not lie in the convex hull of the 4 points. If the convex hull is a convex quadrilateral $ABCD$ and P lies inside this quadrilateral, we let E be the intersection of

AC and BD . WLOG we may assume P lies inside $\triangle ABE$. Then P only lies in $\triangle ABD$ and $\triangle ABC$.

If the convex hull is $\triangle ABC$ and both D, P lie inside this triangle, WLOG we may assume P lies inside $\triangle ABD$. Then P only lies in $\triangle ABD$ and $\triangle ABC$. In any case, there is an even number of triangles containing P .



Now, we consider the general case. We count the number of pairs $(\triangle, \square)$ such that $\triangle$ is a triangle $\triangle A_i A_j A_k$ containing P in its interior and $\square$ is a quadrilateral 3 of whose vertices determine $\triangle$.

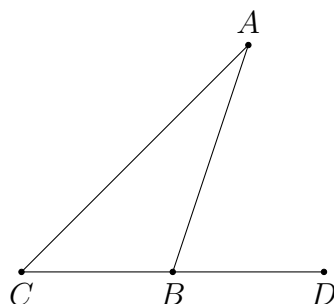
For each $\triangle$ containing P , there are $4n - 3$ choices for the additional vertex to form $\square$. So there are $(4n - 3)t$ pairs in total where t is the number of triangles containing P .

For each $\square$, there is an even number of $\triangle$ such that $(\triangle, \square)$ is counted from above. So the number of pairs is even. As $4n - 3$ is odd, we know that t is even.

Extremal Case

33. Solution 1:

Suppose on the contrary that S is finite. Let A and B be two points in S for which AB is maximized. Let B be the midpoint of CD where $C, D \in S$. Clearly, $C, D \neq A$. WLOG assume $\angle ABC \geq 90^\circ$. Then $\angle ABC > \angle BCA$ and hence $AC > AB$. This is a contradiction. So S must be infinite.



Solution 2:

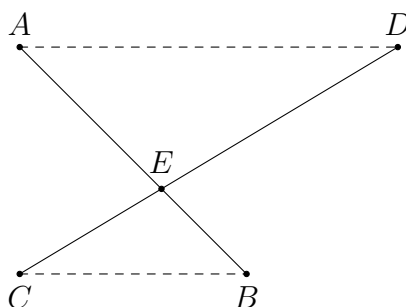
Suppose on the contrary that S is finite. Consider the convex hull of S . Let A be a vertex of this convex hull. Suppose A is the midpoint of BC where $B, C \in S$. Then BAC is a straight line and hence at least one of B and C lies outside the convex hull. This is a contradiction. So S must be infinite.

34. There are finitely many ways to draw n segments such that the endpoints of each segment have distinct colours. We choose such a way for which the sum of the total length of the segments is minimized. We prove that the remaining condition is satisfied. Suppose on the contrary that there are two segments AB and CD which meet each other where A and C are red points, while B and D are blue points.

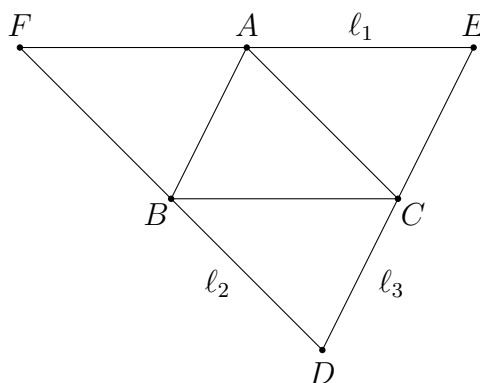
Let AB meet CD at E . If we draw the segments AD and BC instead of AB and CD , then the total length of the segments is reduced since

$$AD + BC < (AE + DE) + (BE + CE) = (AE + BE) + (CE + DE) = AB + CD.$$

This contradicts the minimality of the total length. So we are done.



35. Let A, B, C be points in S such that $[ABC]$ is maximized. Let ℓ_1 be the line passing through A which is parallel to BC . If there exists a point $X \in S$ which lies on different side of ℓ_1 as B , then the distance from X to BC is greater than that from A to BC . This yields the contradiction $[XBC] > [ABC]$. Therefore, X must lie on the same side of ℓ_1 as B .



Similarly, let ℓ_2 be the line passing through B which is parallel to CA , and let ℓ_3 be the line passing through C which is parallel to AB . Then any point in S must lie inside the triangle $\triangle DEF$ bounded by ℓ_1, ℓ_2, ℓ_3 . Note that $\triangle ABC$ is the median triangle of $\triangle DEF$. So $[DEF] = 4[ABC] \leq 4$. Thus, we can simply take $\triangle DEF$.

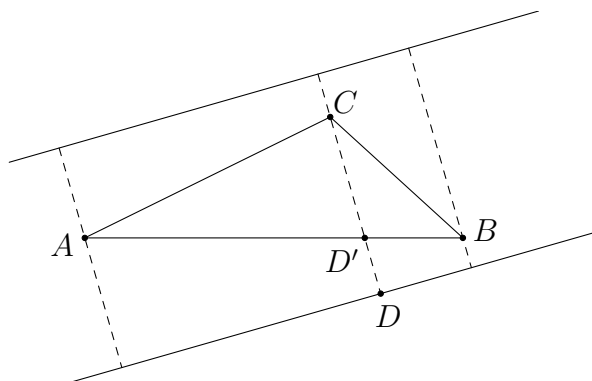
36. Solution 1:

Let A, B, C be points in S such that $[ABC]$ is maximized. Let G be the centroid of $\triangle ABC$. Consider the homothety with centre G and ratio $-\frac{1}{2}$. Then $\triangle ABC$ is mapped to $\triangle DEF$ such that $\triangle ABC$ is the median triangle of $\triangle DEF$. As $\triangle ABC$ can be covered by a strip of width 1, the image of the strip, which has width 2, covers $\triangle DEF$. Since it can be shown that every point in S lies inside $\triangle DEF$, this strip of width 2 covers S as desired.

Solution 2:

Let A and B be two points in S such that the length AB is maximized. We claim that the strip w of width 2 formed by two lines parallel to AB such that AB lies in the middle of the strip contains every point.

Pick any other point C . It is known that $\triangle ABC$ lies in a strip of width 1. We draw the perpendicular lines from A, B, C to that strip. Consider the perpendicular in the middle, say the one from C . Then we can project C to one of the boundaries of the strip to get point D such that the segment CD meets the side AB . Let CD meet AB at D' . Then $CD' \leq CD \leq 1$. It follows that the altitude from C to AB has length at most 1.

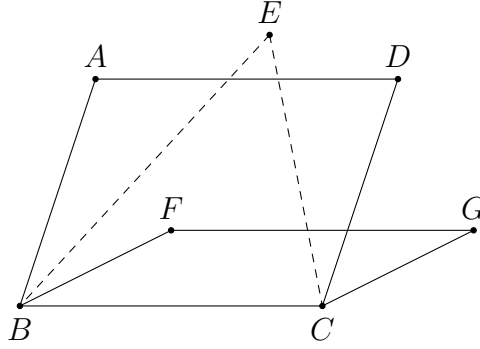


In general, we only know that one of the altitudes of $\triangle ABC$ has length at most 1. But since AB is the longest side, the altitude from C must have the shortest length. So its length cannot exceed 1 in any case. Thus, C lies inside w and we are done.

37. If $|S| < 3$, then the condition is trivially satisfied. If $|S| \geq 3$, we take any 3 points in S . The condition shows that there is a fourth point in S forming a parallelogram with these 3 points. So $|S| \geq 4$. The case $|S| = 4$ with the four points forming a parallelogram is

clearly a solution. If there are 4 points in S not forming a parallelogram, we can choose any 3 of them to show the existence of a fifth point in S . It remains to consider the case $|S| \geq 5$.

Let A, B, C be 3 points in S for which $[ABC]$ is maximized. Let D be a point in S such that $ABCD$ is a parallelogram. Note that $[ABC] = [BCD] = [CDA] = [DAB]$. If there exists a point $E \in S$ lying outside $ABCD$, WLOG we may assume E lies on different side of the line AD as B and C . Then $[EBC] > [ABC]$, which is a contradiction.



So $ABCD$ is the convex hull of S . As $|S| \geq 5$, there exists $F \in S$. WLOG assume F lies inside $ABCD$ or on the boundary AD . Consider the points B, C, F . There exists $G \in S$ such that $BCGF$, $BCFG$ or $BGCF$ is a parallelogram. In any case, G must lie outside $ABCD$, which is a contradiction.

Therefore, S can be any set with at most 2 points, or a set of 4 points forming a parallelogram.

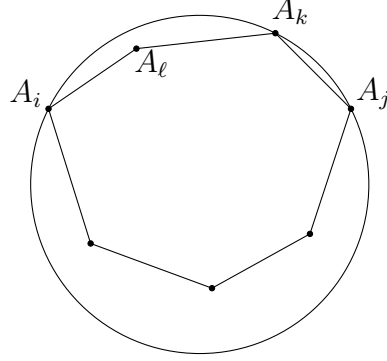
38. Let $\triangle A_i A_j A_k$ be a triangle with largest circumradius. WLOG assume both $\angle A_k A_i A_j$ and $\angle A_k A_j A_i$ are acute. Consider any other vertex A_ℓ . WLOG assume A_ℓ lies on the same side of $A_j A_k$ as A_i . Since the circumradius of $\triangle A_j A_k A_\ell$ does not exceed that of $\triangle A_i A_j A_k$, we have

$$\frac{A_j A_k}{\sin \angle A_j A_\ell A_k} \leq \frac{A_j A_k}{\sin \angle A_j A_i A_k}.$$

This implies $\sin \angle A_j A_i A_k \leq \sin \angle A_j A_\ell A_k$, and hence $\angle A_j A_i A_k \leq \angle A_j A_\ell A_k$ as $\angle A_j A_i A_k$ is acute. This implies A_ℓ lies inside $(A_i A_j A_k)$. Thus, the polygon lies strictly inside $(A_i A_j A_k)$.

Suppose there exists a point A_ℓ lying on the same side of $A_i A_j$ as A_k . WLOG assume $A_i A_j A_k A_\ell$ is a convex quadrilateral. As A_ℓ lies inside $(A_i A_j A_k)$, we have

$$\angle A_k A_j A_i + \angle A_i A_\ell A_k \geq 180^\circ.$$



Since $\angle A_k A_j A_i$ is acute, this implies $\sin \angle A_i A_l A_k \leq \sin \angle A_k A_j A_i$. Thus,

$$\frac{A_i A_k}{\sin \angle A_k A_j A_i} \leq \frac{A_i A_k}{\sin \angle A_i A_l A_k}.$$

This shows the radius of $(A_i A_j A_k)$ is at most the radius of $(A_i A_k A_l)$. From the choice of the points, these radii must be equal. Then we can replace A_i by A_l and work on $\triangle A_j A_k A_l$. Repeating the process, we can end up with 3 consecutive vertices whose circumradius is the greatest possible.